

YAYLYU, Russian Federation

WMO: 360640

Lat: 51.7715N

Lon: 87.6044E

Elev: 480

StdP: 95.69

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-23.7	-20.7	-27.8	0.3	-23.5	-24.6	0.4	-20.3	7.3	-3.6	6.7	-3.9	2.4	320	0.555

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.8	27.6	17.9	25.6	17.3	23.7	16.6	19.1	25.5	18.2	23.7	17.4	22.2	2.4	350

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.0	12.9	20.9	16.2	12.2	19.9	15.5	11.7	19.3	56.6	25.4	53.5	23.6	50.9	22.3	23.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.0	5.2	4.4	DB	-25.6	31.7	4.0	1.3	-28.5	32.7	-30.8	33.5	-33.1	34.2	-36.0	35.2	(4)
(5)			WB	-25.8	21.4	3.9	1.0	-28.7	22.2	-31.0	22.8	-33.2	23.4	-36.1	24.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	4.6	-9.0	-8.3	-0.8	5.8	9.4	15.9	17.6	16.4	10.9	4.9	-2.4	-6.3	(6)
(7)		DBStd	10.39	5.62	5.61	5.09	4.69	4.10	3.62	2.78	2.85	3.86	4.32	5.48	5.44	(7)
(8)		HDD10.0	2715	590	512	334	137	62	2	0	1	35	166	372	505	(8)
(9)		HDD18.3	5072	848	745	592	376	277	88	46	72	224	417	622	763	(9)
(10)		CDD10.0	740	0	0	0	12	44	180	237	199	61	7	0	0	(10)
(11)		CDD18.3	54	0	0	0	0	1	16	24	13	1	0	0	0	(11)
(12)		CDH23.3	478	0	0	0	0	11	189	190	82	7	0	0	0	(12)
(13)		CDH26.7	92	0	0	0	0	1	42	38	10	1	0	0	0	(13)
(14)	Wind	WSAvg	1.8	2.5	2.0	1.8	1.6	1.5	1.4	1.4	1.5	1.6	1.8	2.2	2.7	(14)
(15)	Precipitation	PrecAvg	845	18	18	29	78	107	109	133	114	88	68	51	31	(15)
(16)		PrecMax	1096	38	40	81	166	225	199	224	240	162	196	103	66	(16)
(17)		PrecMin	619	6	2	4	33	47	56	65	58	30	26	14	10	(17)
(18)		PrecStd	104	9	9	17	39	45	35	41	44	41	44	26	15	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	3.8	5.6	13.3	21.3	25.0	30.1	29.9	28.1	24.2	17.2	12.0	5.7	(19)
(20)			MCWB	-0.4	1.5	5.6	10.5	13.8	18.5	19.6	18.7	15.5	9.7	6.3	0.7	(20)
(21)		2%	DB	1.4	2.8	10.5	18.3	21.7	27.7	27.6	25.5	21.4	14.8	8.3	3.8	(21)
(22)			MCWB	-1.8	-0.4	4.0	9.1	12.0	17.3	18.5	17.5	14.0	8.6	3.9	-0.1	(22)
(23)		5%	DB	-0.4	1.1	8.5	15.7	19.1	25.6	25.5	23.6	19.3	13.2	6.2	1.8	(23)
(24)			MCWB	-3.3	-1.8	2.9	7.9	11.0	16.8	17.8	16.8	12.7	7.5	2.4	-0.9	(24)
(25)		10%	DB	-1.8	-0.6	6.4	13.2	16.7	23.2	23.5	22.0	17.2	11.2	4.4	0.3	(25)
(26)			MCWB	-4.2	-3.4	1.7	6.9	10.0	16.0	17.4	16.0	11.6	6.5	1.7	-2.3	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.2	1.9	6.3	11.5	15.0	19.6	20.9	19.6	16.4	10.7	6.7	1.7	(27)
(28)			MCDWB	2.8	5.1	12.7	19.6	21.9	27.4	27.6	26.1	22.6	15.0	11.3	4.2	(28)
(29)		2%	WB	-1.7	0.2	4.5	10.0	13.1	18.5	19.5	18.3	14.6	9.3	4.2	0.3	(29)
(30)			MCDWB	1.2	2.5	9.7	16.5	19.1	25.5	25.5	23.9	20.3	13.7	7.0	3.1	(30)
(31)		5%	WB	-3.0	-1.8	3.2	8.6	11.9	17.5	18.6	17.5	13.4	8.0	3.0	-0.8	(31)
(32)			MCDWB	-0.7	0.9	7.5	14.0	17.0	23.4	23.9	22.3	18.1	12.4	5.5	2.0	(32)
(33)		10%	WB	-4.3	-3.3	2.0	7.4	10.8	16.6	17.7	16.6	12.3	6.8	1.7	-2.2	(33)
(34)			MCDWB	-1.9	-0.9	5.7	11.9	15.6	21.8	22.2	20.7	16.0	10.7	4.2	0.1	(34)
(35)	Mean Daily Temperature Range	MDBR		6.7	8.3	9.2	10.2	10.7	11.6	9.8	9.4	9.2	7.2	6.0	5.8	(35)
(36)		5% DB	MCDBR	6.3	8.3	11.3	15.3	15.5	16.5	14.5	12.8	12.3	10.0	7.1	6.0	(36)
(37)			MCWBR	4.2	5.6	6.4	7.7	7.9	7.4	6.5	6.0	6.0	5.0	4.3	3.8	(37)
(38)			MCDBR	5.9	8.2	10.3	12.6	12.7	13.7	11.8	10.8	10.9	9.2	6.9	6.2	(38)
(39)		5% WB	MCWBR	4.2	5.8	6.2	7.0	7.0	6.7	5.9	5.5	5.8	5.0	4.5	4.3	(39)
(40)	Clear-Sky Solar Irradiance	taub		0.292	0.331	0.389	0.447	0.431	0.421	0.432	0.417	0.376	0.367	0.331	0.296	(40)
(41)		taud		2.054	1.990	1.924	1.970	2.129	2.272	2.267	2.273	2.317	2.197	2.075	2.016	(41)
(42)		Ebn at Noon		675	751	776	782	823	835	818	809	801	712	613	584	(42)
(43)		Edh at Noon		80	115	150	163	145	128	126	119	101	93	79	71	(43)
(44)	All-Sky Solar Radiation	RadAvg		1.27	2.32	3.71	4.51	5.20	5.84	5.41	4.67	3.49	1.97	1.12	0.89	(44)
(45)		RadStd		0.13	0.15	0.32	0.34	0.53	0.53	0.32	0.34	0.28	0.23	0.12	0.10	(45)

Historical Trends

		DBAvg		Heating		Cooling		Degree-Days			
				99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (4 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page